

Southeastern Section - 68th Annual Meeting - 2019

Paper No. 16-8

Presentation Time: 1:00 PM-5:00 PM

ARTIFICIAL NEURAL NETWORKS APPLICATION IN TIME SERIES WATER DISCHARGE FOR THE PREDICTION OF FLOODING IN MOHAWK RIVER, NEW YORK

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A time series model enhanced with an artificial neural network model has been used for the prediction of flood events in Mohawk River, New York. For the prediction of the water discharge time series, climatic and hydrogeological variables have been used in the analysis. To separate the different signals in the time series a low pass filter has been applied in the raw data. The Kolmogorov-Zurbenko filter has been used for the decomposition of the time series into the long-, seasonal-, and short-term component. Each time series component has been modeled using the multiple linear regression model (MLR), and the artificial neural network (ANN) model. The time series decomposition combined with the ANN model provide the flood prediction of the water discharge time series and lead to a forecasting up to 96% when the climatic and hydrogeological variables have been used in the model. Although the MLR model provides a lower performance, it retains the advantage of the physical interpretation of the water discharge time series, while the ANN model shows as an outstanding prediction model.

Session No. 16--Booth# 8

[T12. Natural Hazards Prediction \(Posters\)](#)

Thursday, 28 March 2019: 1:00 PM-5:00 PM

[Carolina Ballroom \(Francis Marion Hotel\)](#)

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